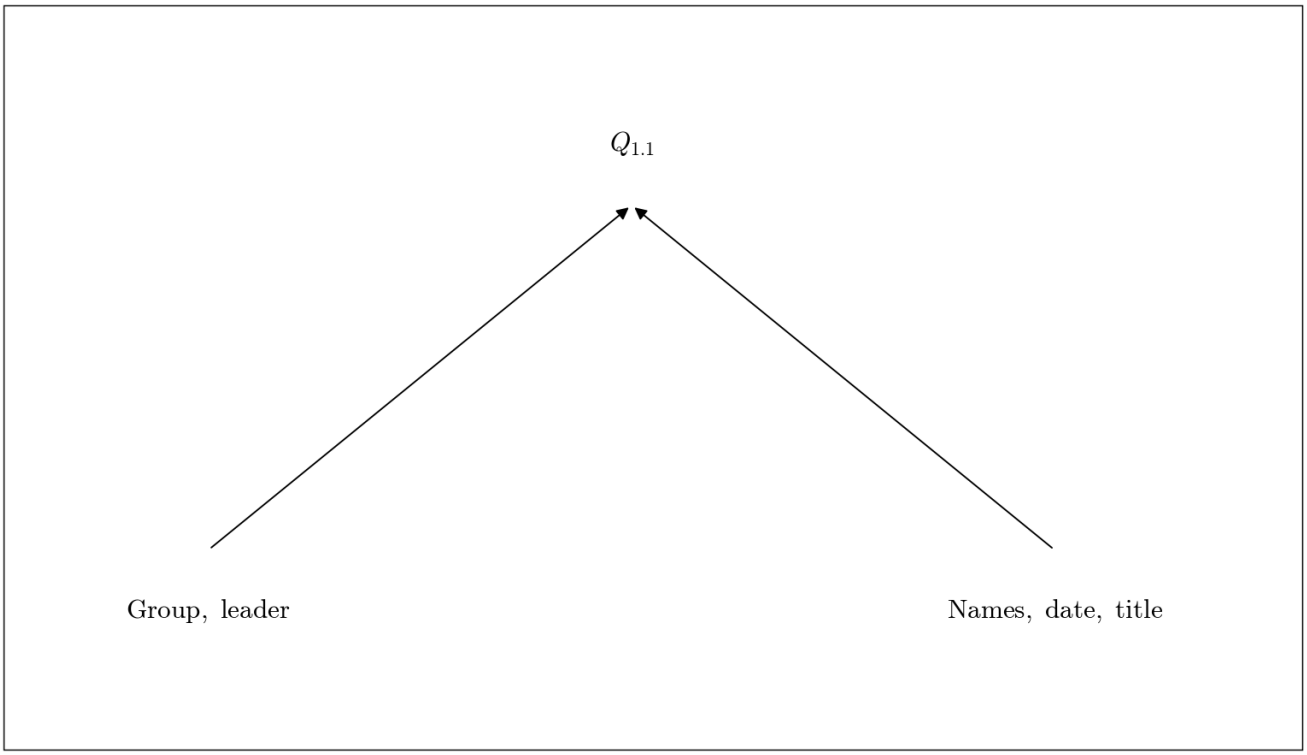
Q1.1 Group information

All three members confirm the group number, leader, member names, and experiment date.

Deliverables

Enter the group number, leader, all member names, date, and full lab title. Leave unknown information pending confirmation.



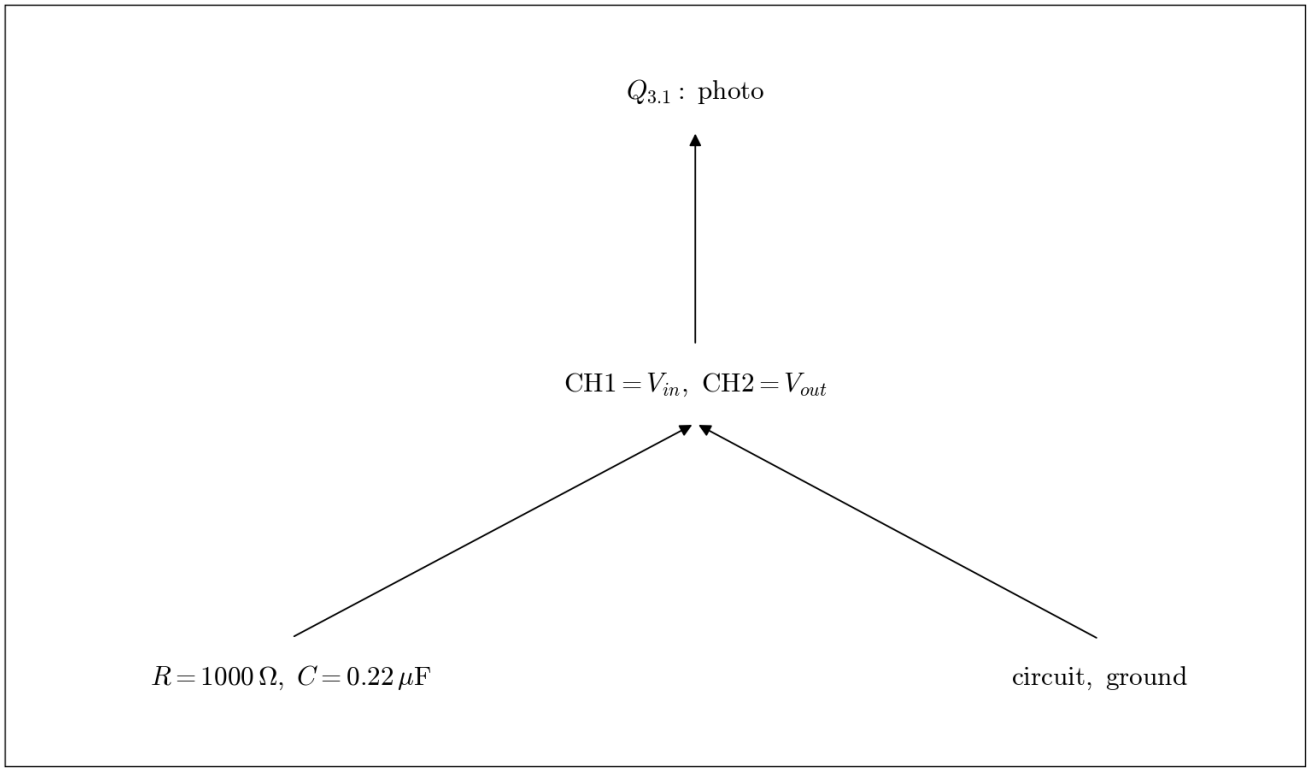
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q3.1 RC Setup photograph

Wire the circuit shown in the instructions. Connect CH1 to Vin and CH2 to Vout; verify the common ground.

Deliverables

Provide an actual breadboard photograph showing components, input, output, and ground connections. An oscilloscope screenshot does not replace this photograph.



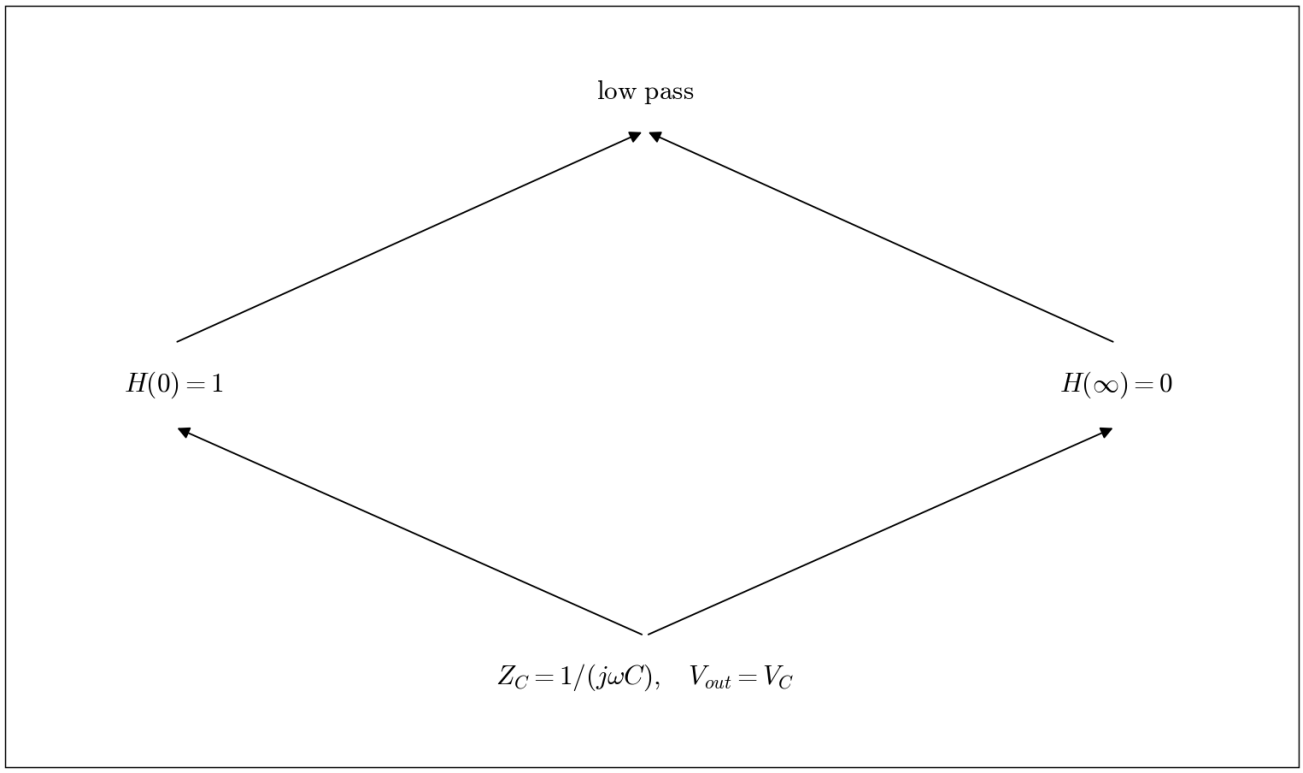
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q3.2 RC Filter classification

Locate the output element, then evaluate the low-frequency and high-frequency limits.

Deliverables

Low-pass: at low frequency the capacitor approaches an open circuit and Vout approaches Vin;
at high frequency it approaches a short circuit and Vout approaches zero.



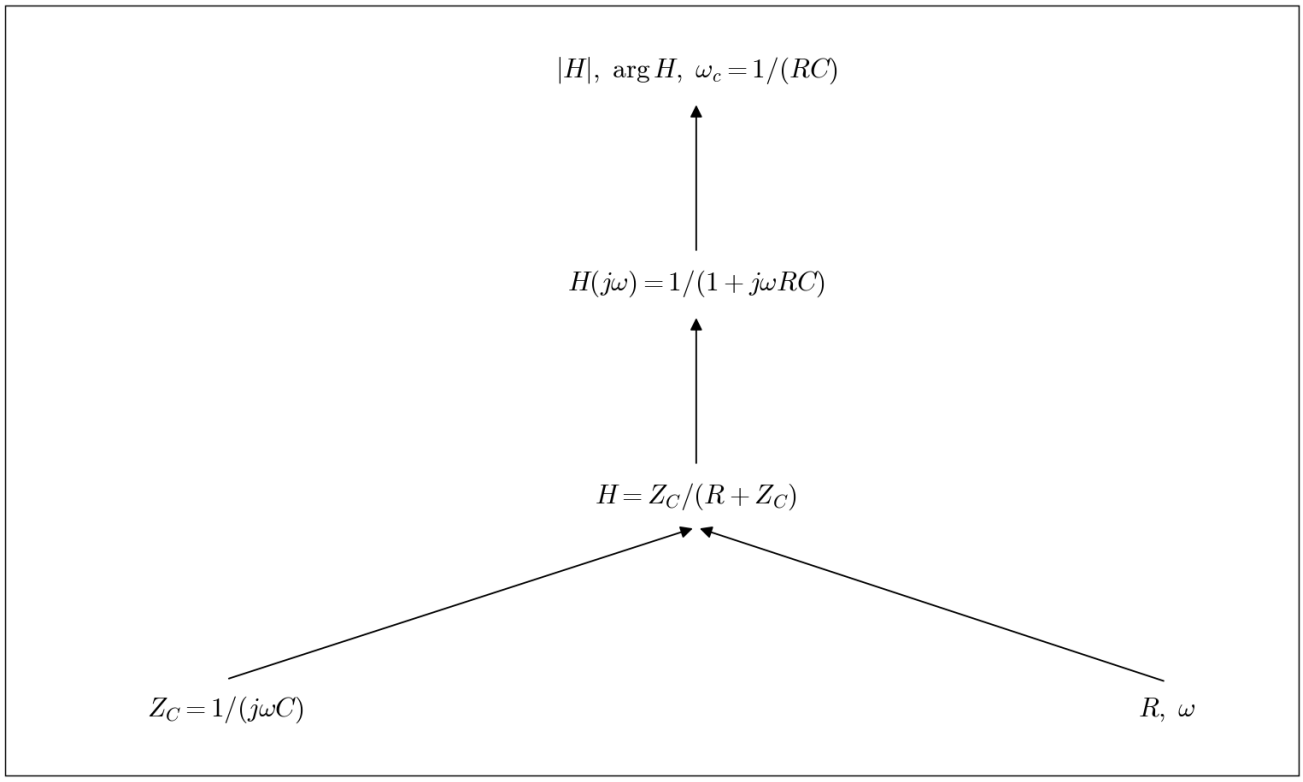
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q3.3 RC Transfer function derivation

Start with the series voltage divider. Derive H, the cutoff angular frequency, the magnitude, and the phase.

Deliverables

Show the voltage divider, algebraic simplification, |H|, arg H, and omega_c.



Arrows point toward dependent results. Planning guide; experimental work remains pending.

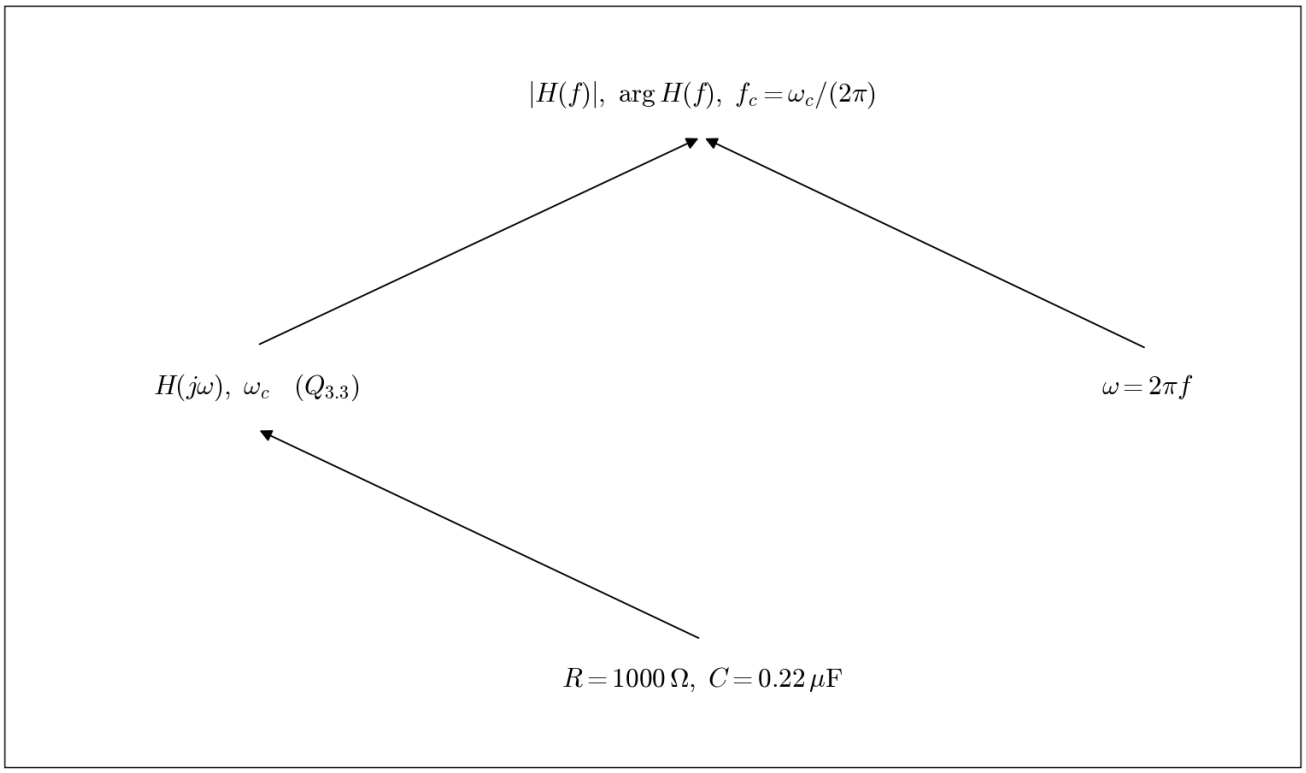
Q3.4 RC Numerical expressions in Hz

Use Q3.3 and the component values in the original schematic. Substitute omega = 2 pi f.

Deliverables

Provide numerical expressions for magnitude and phase, and the ideal cutoff frequency:

723.43 Hz.



Arrows point toward dependent results. Planning guide; experimental work remains pending.

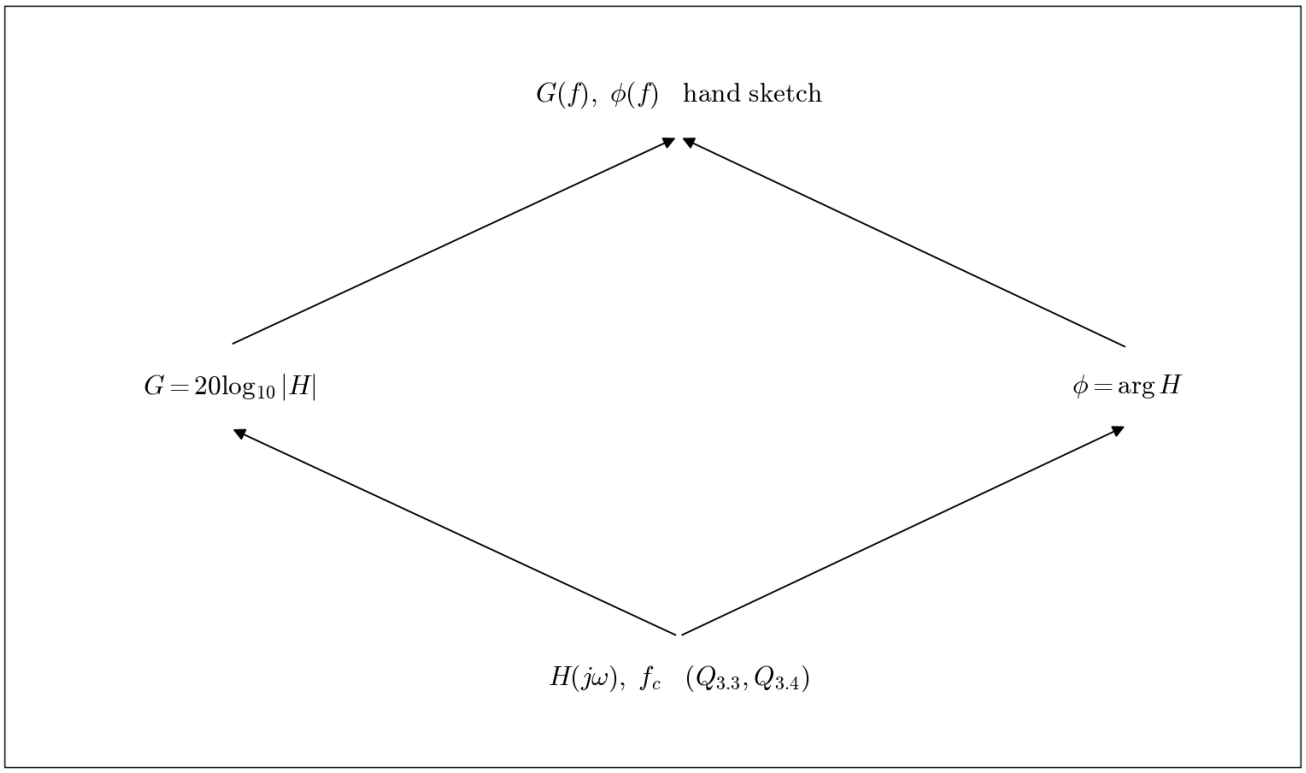
Q3.5 RC Hand-drawn Bode plots

Use Q3.3 and the cutoff frequency from Q3.4 to label the frequency axis in Hz.

Deliverables

Draw magnitude and phase by hand. Label fc, -3.01 dB, the cutoff phase, and asymptotes.

Include your own hand-drawn work.



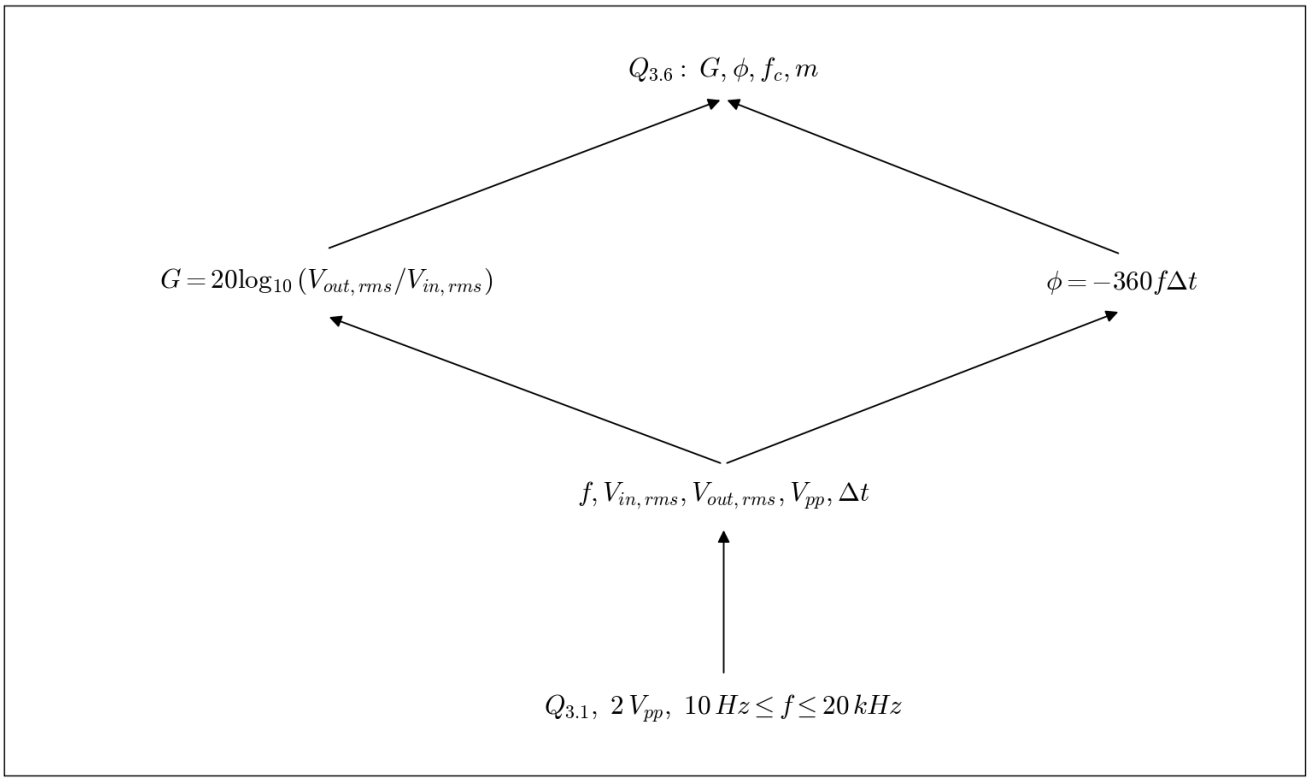
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q3.6 RC Measured frequency sweep

Complete the setup in Q3.1. Start with a 2 Vpp, 1 kHz sine wave, then measure all 11 required frequencies. Add points near cutoff.

Deliverables

For every point, record actual frequency, Vin/Vout in VRMS and Vpp, cursor time difference, phase, and gain. Save a full-screen capture with measurements and cursors visible. Report the measured half-power frequency and stopband slope. Place RC captures in Appendix B.



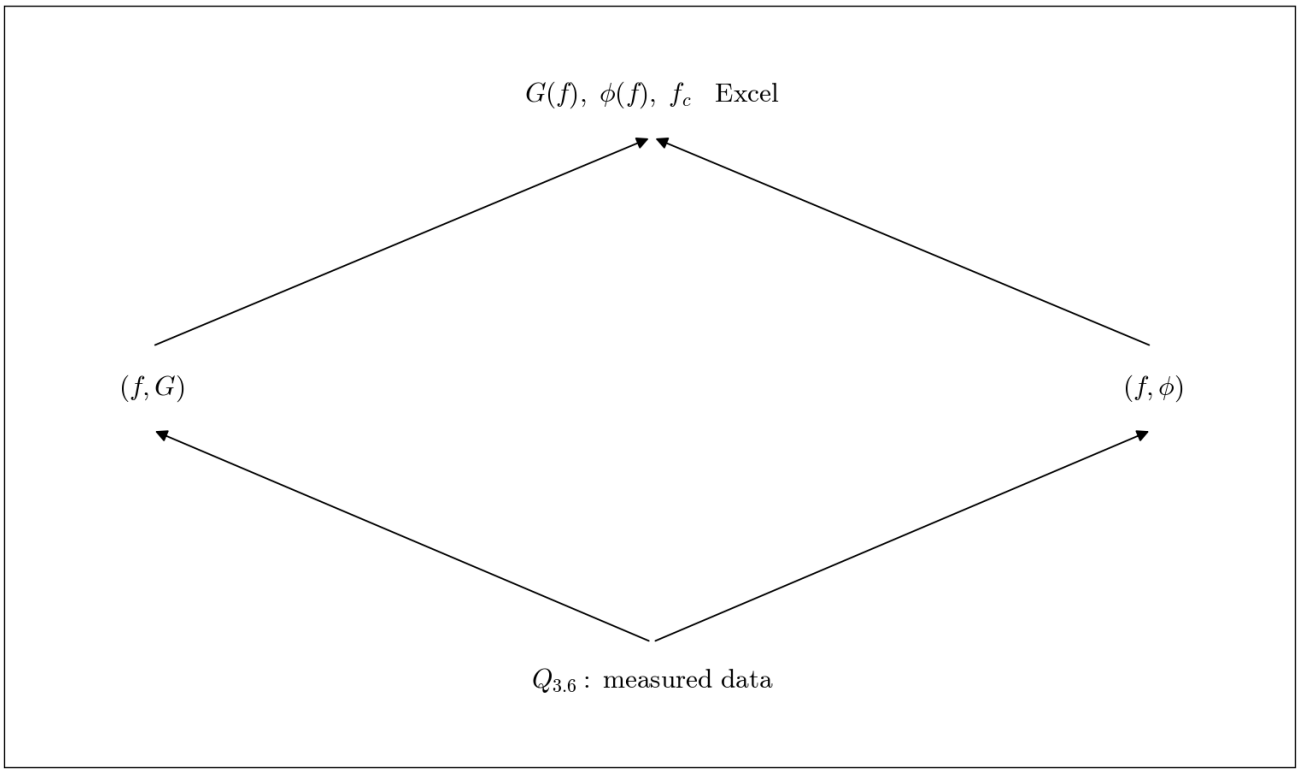
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q3.7 RC Experimental Bode plots in Excel

Complete the measured data in Q3.6 first. Use experimental values for these plots.

Deliverables

Use Excel to create gain in dB and phase in degrees versus frequency. Use a logarithmic numeric frequency axis and label the measured cutoff.



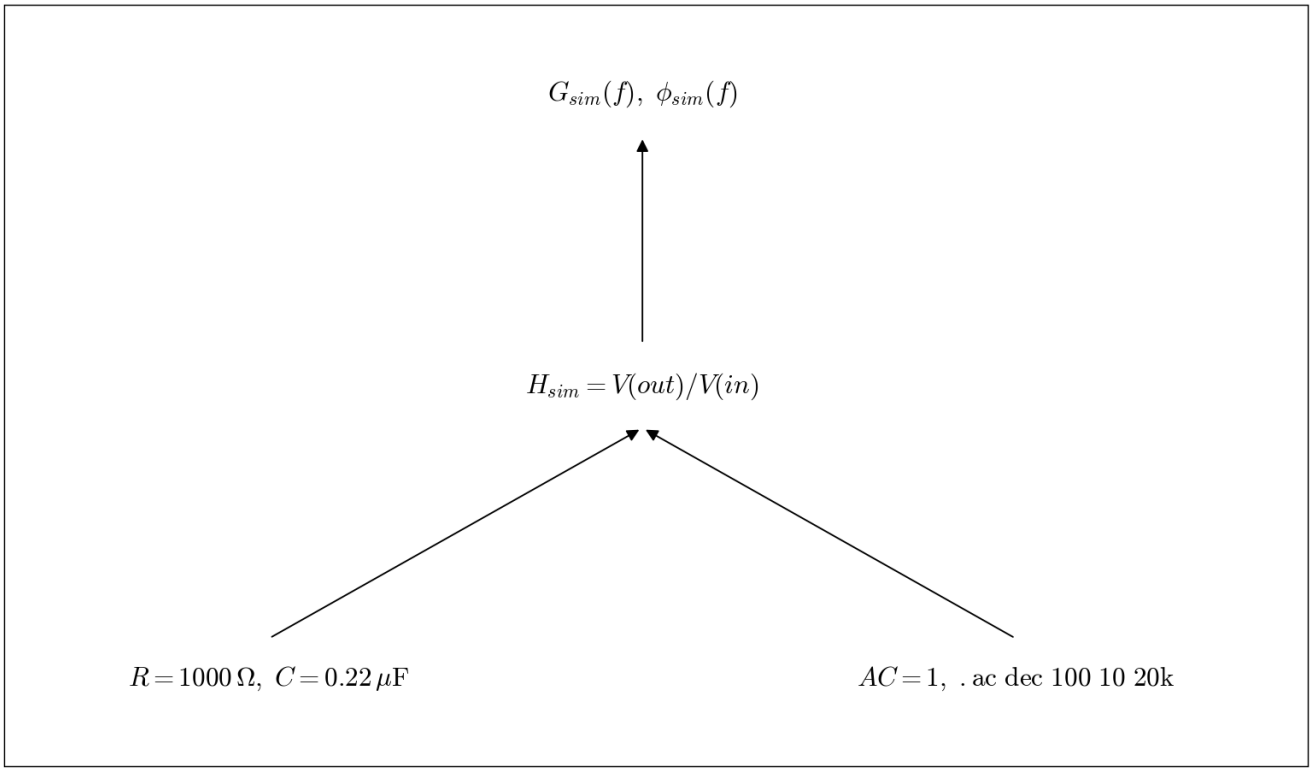
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q3.8 RC LTspice AC simulation

Build a separate simulation from the original schematic. This can proceed in parallel with measurements; verify matching component values.

Deliverables

Use a source with AC amplitude 1 and .ac dec 100 10 20k. Display dB(V(out)/V(in)) and the phase of V(out)/V(in). Save the circuit, settings, and result captures.



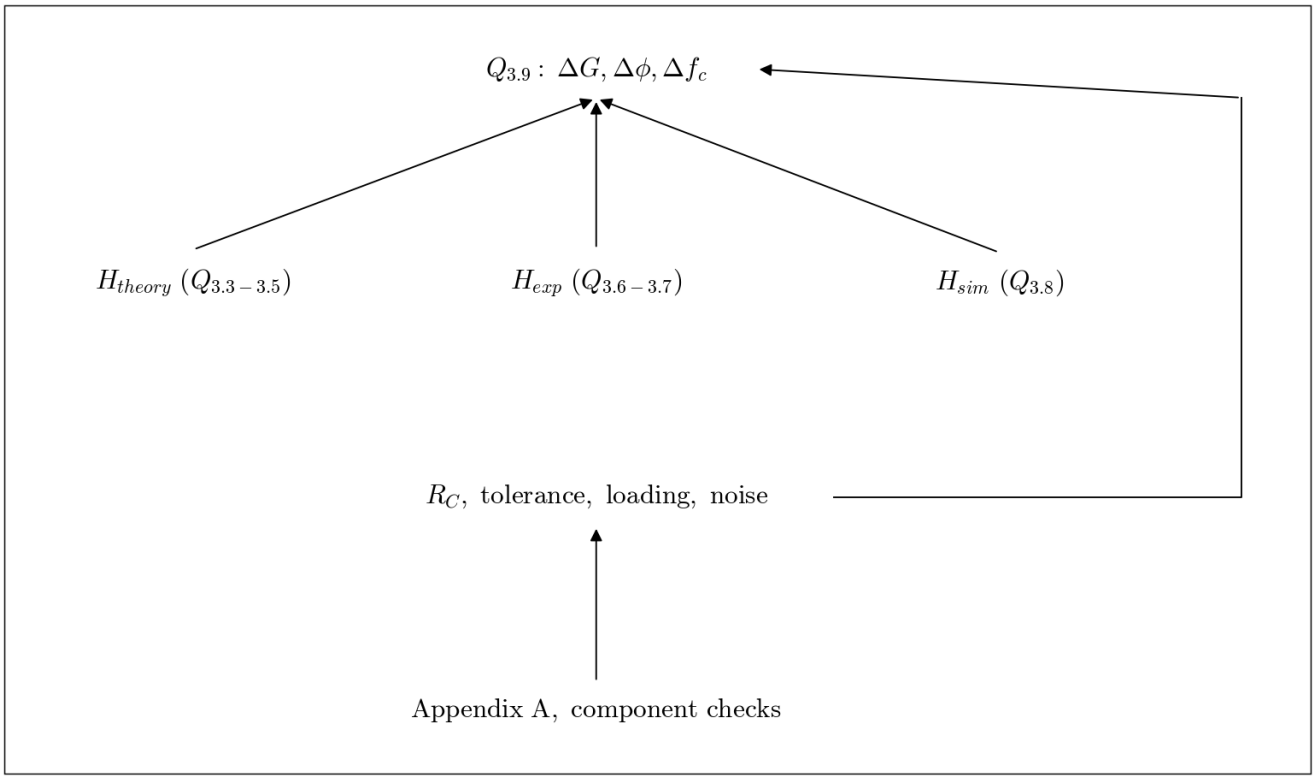
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q3.9 RC Theory simulation and measurement comparison

Combine Q3.3 through Q3.8. Compare magnitude, phase, cutoff frequency, and stopband trends.

Deliverables

Compare all three results and explain differences. The instructions explicitly require theoretical and measured magnitude curves on the same Excel chart. Also check phase. Treat proposed nonideal effects as hypotheses supported by evidence.



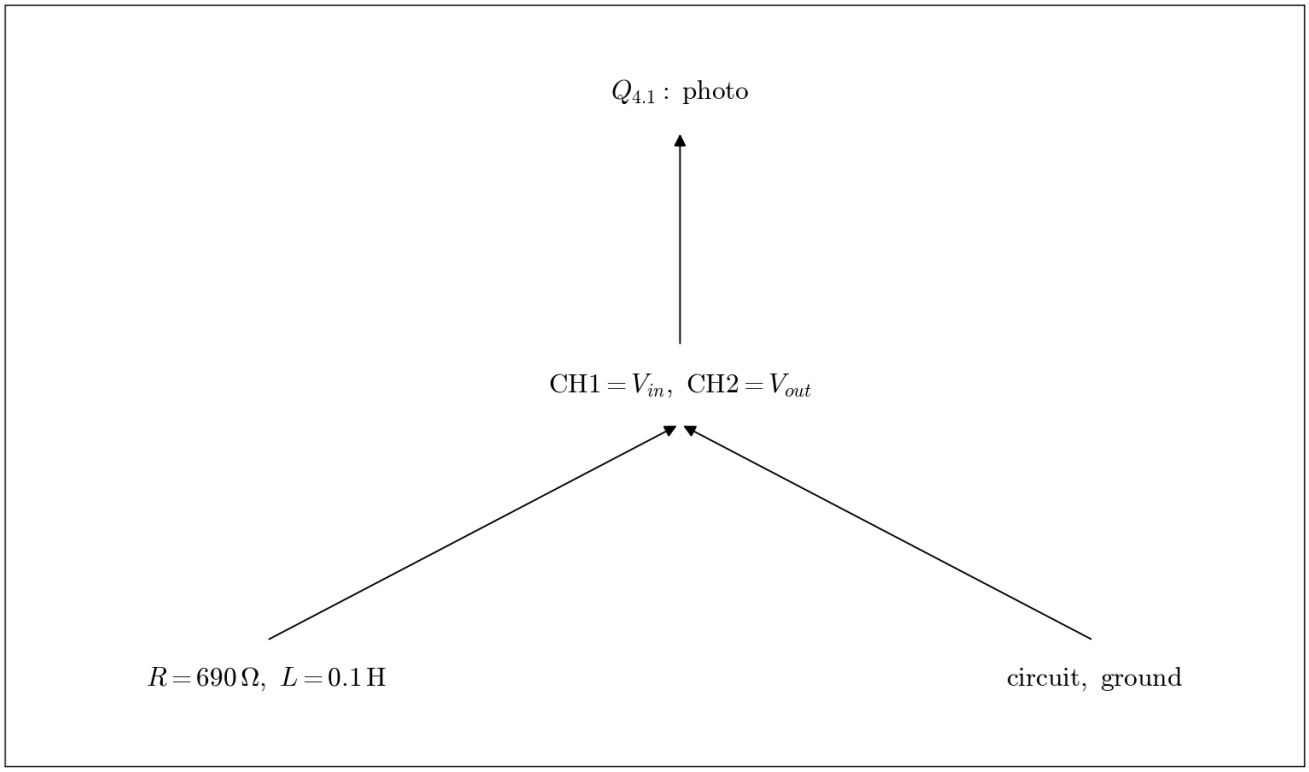
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q4.1 RL Setup photograph

Wire the circuit shown in the instructions. Connect CH1 to Vin and CH2 to Vout; verify the common ground.

Deliverables

Provide an actual breadboard photograph showing components, input, output, and ground connections. An oscilloscope screenshot does not replace this photograph.



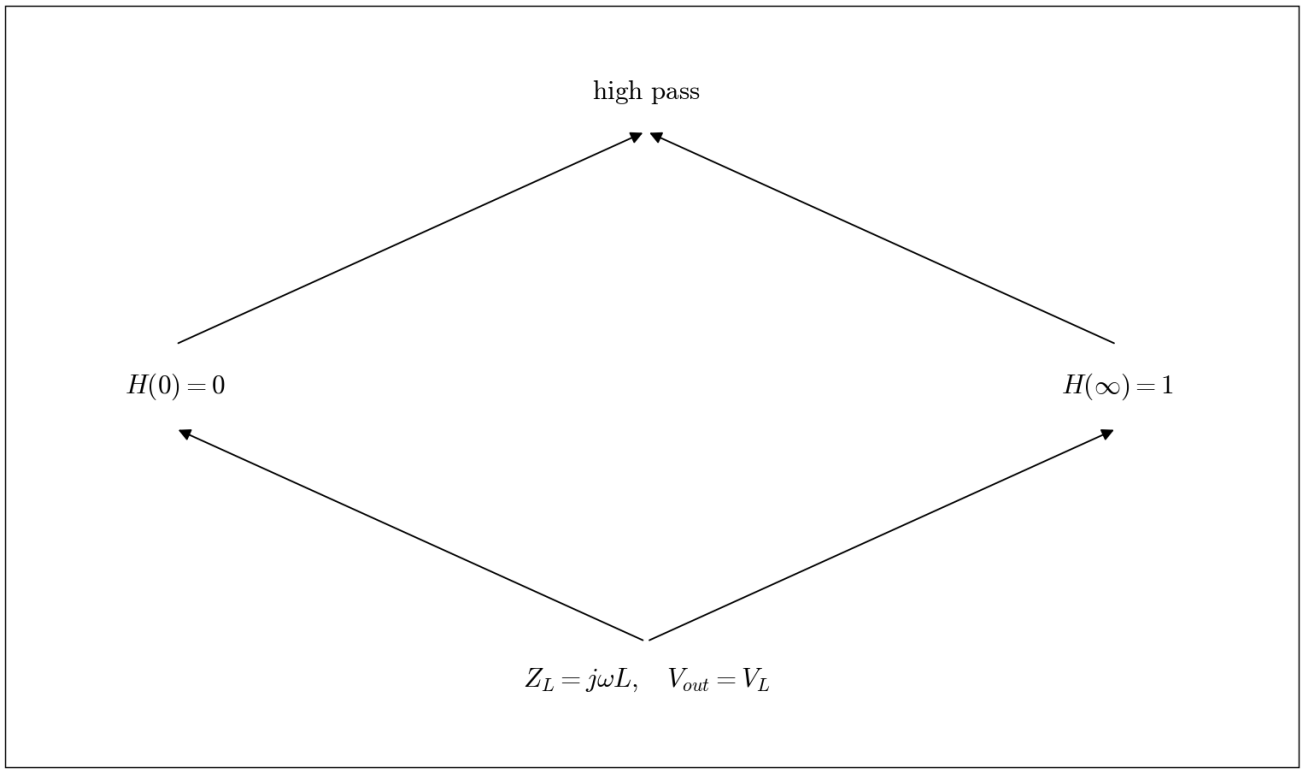
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q4.2 RL Filter classification

Locate the output element, then evaluate the low-frequency and high-frequency limits.

Deliverables

High-pass: at low frequency the inductor approaches a short circuit and Vout approaches zero; at high frequency its impedance increases and Vout approaches Vin.



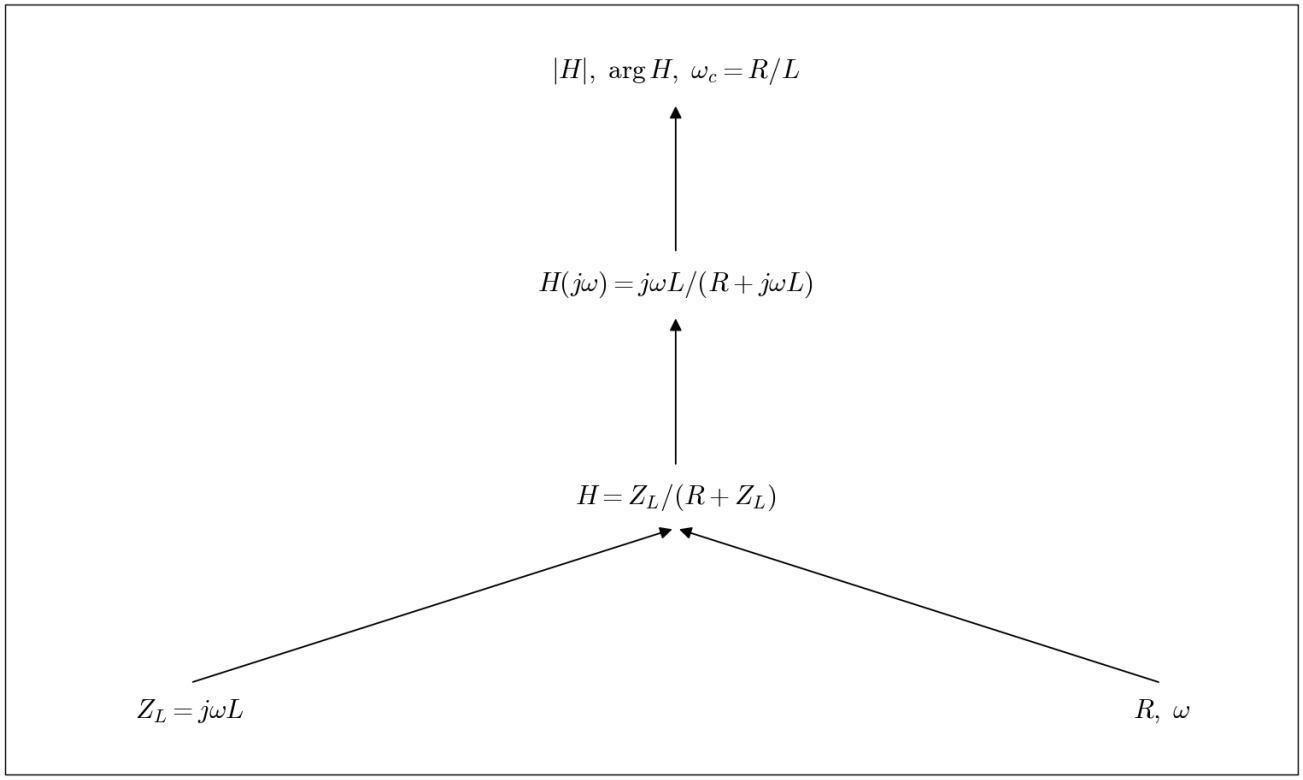
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q4.3 RL Transfer function derivation

Start with the series voltage divider. Derive H, the cutoff angular frequency, the magnitude, and the phase.

Deliverables

Show the voltage divider, algebraic simplification, |H|, arg H, and omega_c. The references to capacitor C in the question are typos; use inductor L.



Arrows point toward dependent results. Planning guide; experimental work remains pending.

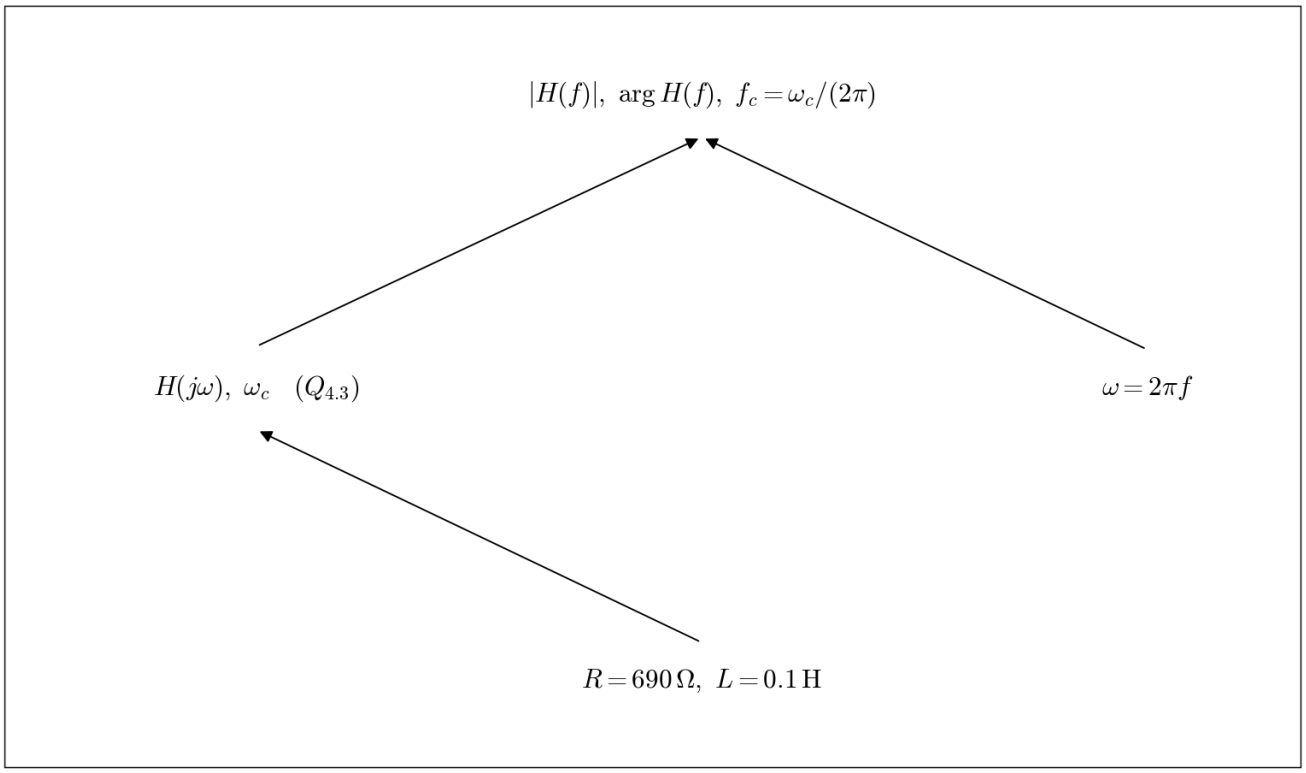
Q4.4 RL Numerical expressions in Hz

Use Q4.3 and the component values in the original schematic. Substitute omega = 2 pi f.

Deliverables

Provide numerical expressions for magnitude and phase, and the ideal cutoff frequency:

1098.17 Hz. The question should refer to R and L.



Arrows point toward dependent results. Planning guide; experimental work remains pending.

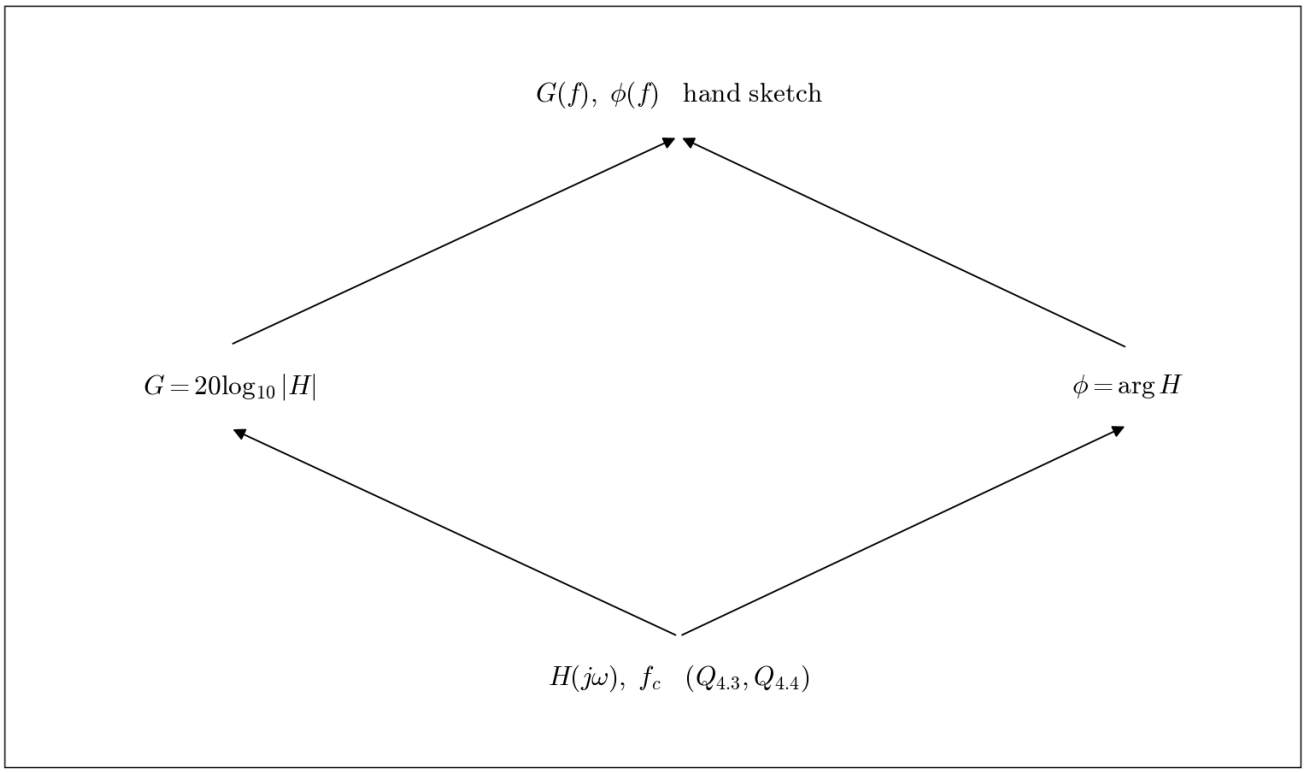
Q4.5 RL Hand-drawn Bode plots

Use Q4.3 and the cutoff frequency from Q4.4 to label the frequency axis in Hz.

Deliverables

Draw magnitude and phase by hand. Label fc, -3.01 dB, the cutoff phase, and asymptotes.

Include your own hand-drawn work.



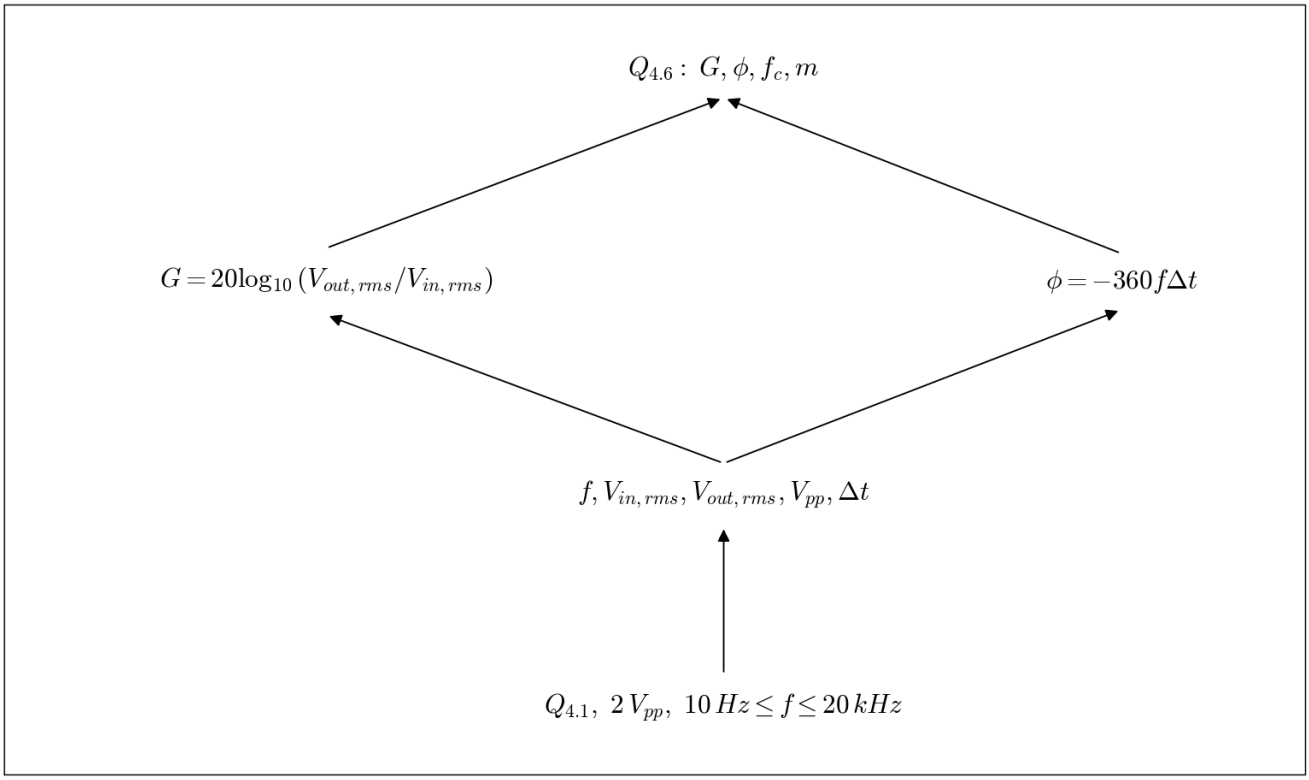
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q4.6 RL Measured frequency sweep

Complete the setup in Q4.1. Start with a 2 Vpp, 1 kHz sine wave, then measure all 11 required frequencies. Add points near cutoff.

Deliverables

For every point, record actual frequency, Vin/Vout in VRMS and Vpp, cursor time difference, phase, and gain. Save a full-screen capture with measurements and cursors visible. Report the measured half-power frequency and stopband slope. Place RL captures in Appendix C; Q4.6 says Appendix B, which conflicts with the final appendix heading.



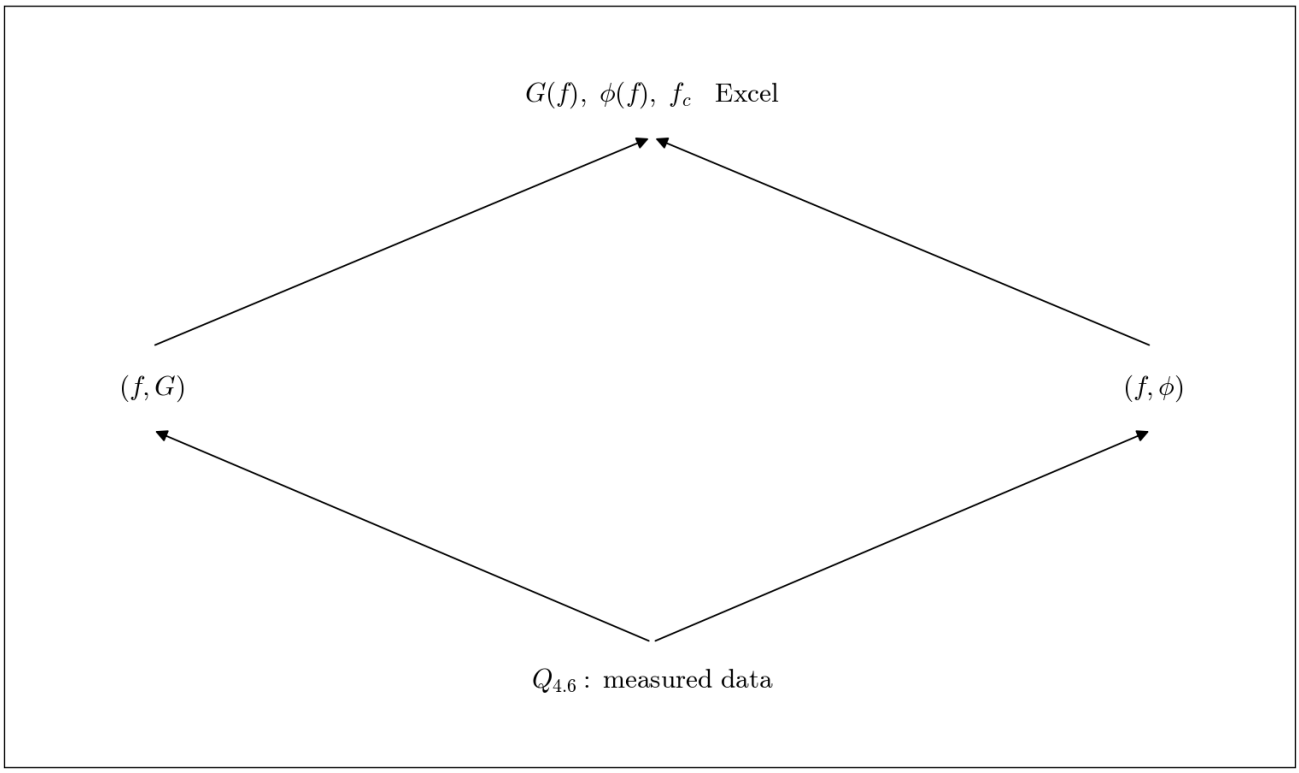
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q4.7 RL Experimental Bode plots in Excel

Complete the measured data in Q4.6 first. Use experimental values for these plots.

Deliverables

Use Excel to create gain in dB and phase in degrees versus frequency. Use a logarithmic numeric frequency axis and label the measured cutoff.



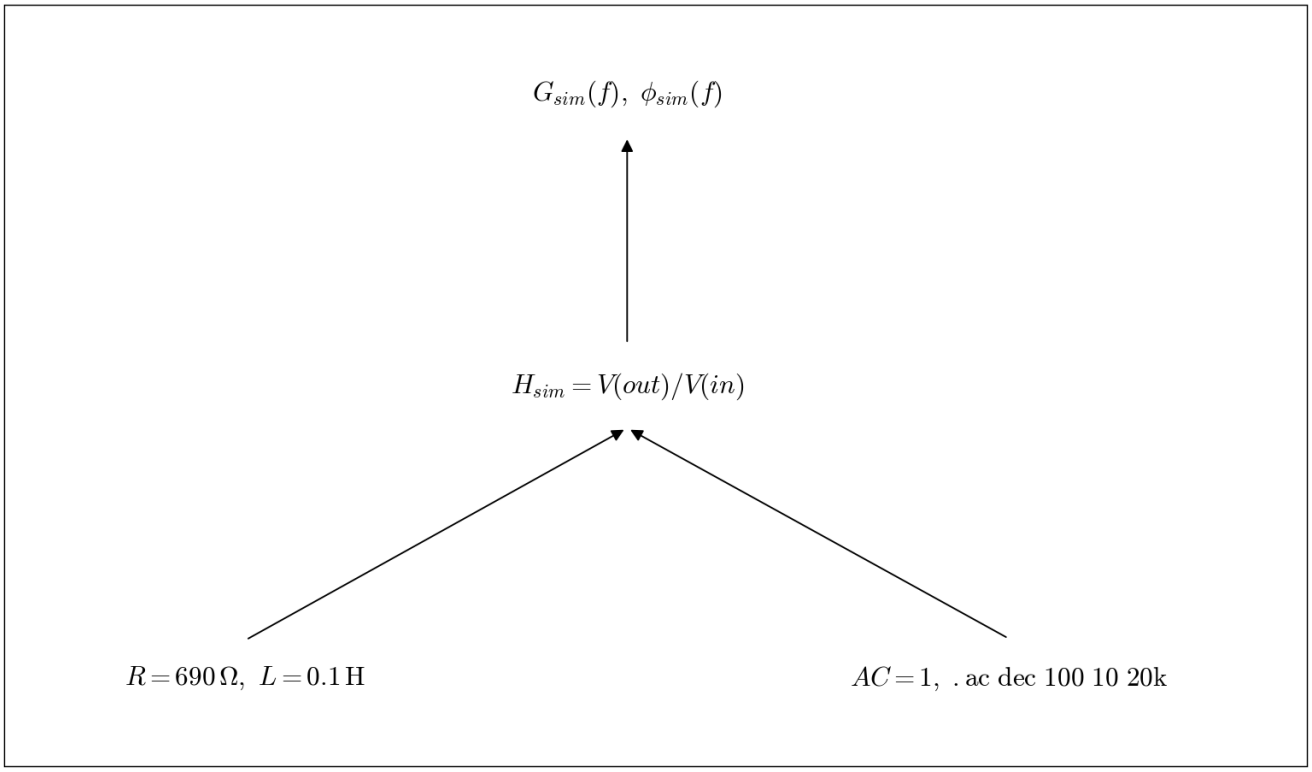
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q4.8 RL LTspice AC simulation

Build a separate simulation from the original schematic. This can proceed in parallel with measurements; verify matching component values.

Deliverables

Use a source with AC amplitude 1 and .ac dec 100 10 20k. Display dB(V(out)/V(in)) and the phase of V(out)/V(in). Save the circuit, settings, and result captures.



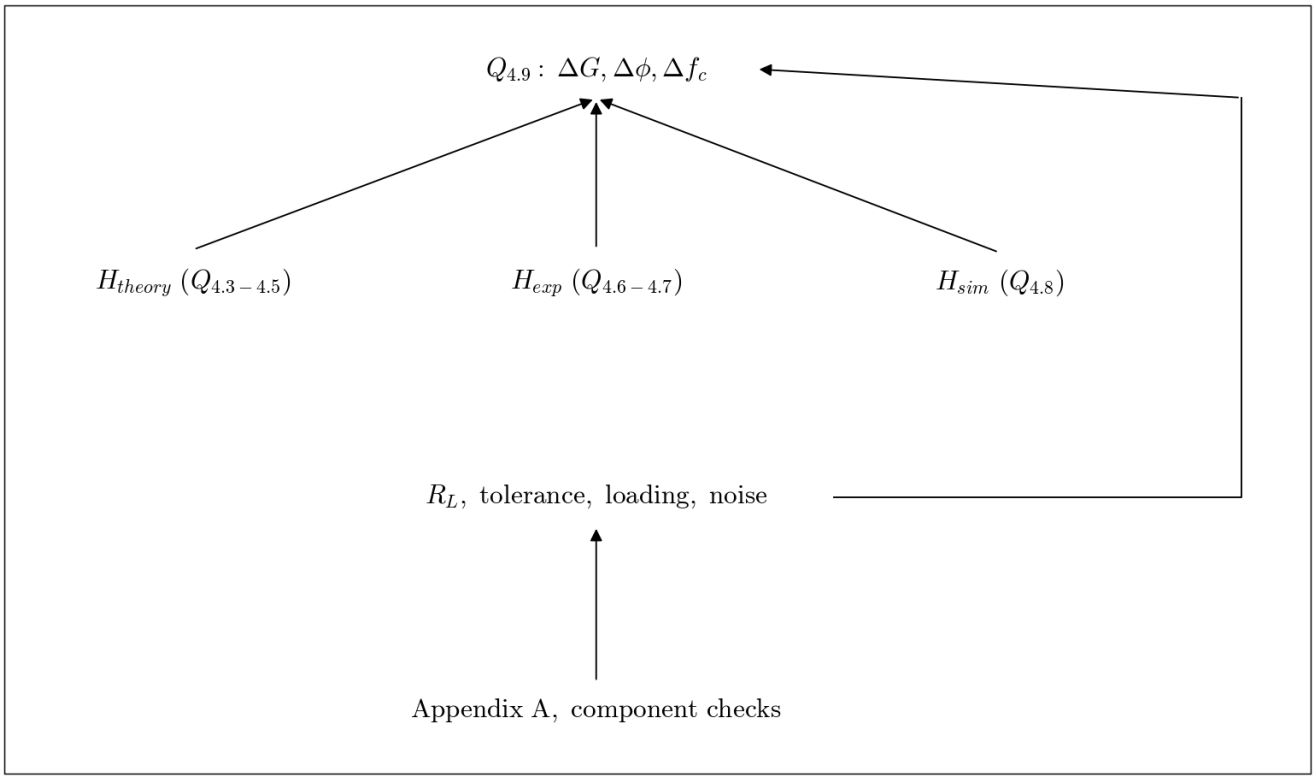
Arrows point toward dependent results. Planning guide; experimental work remains pending.

Q4.9 RL Theory simulation and measurement comparison

Combine Q4.3 through Q4.8. Compare magnitude, phase, cutoff frequency, and stopband trends.

Deliverables

Compare all three results and explain differences. Compare the three Bode plots. An Excel overlay is recommended for clarity. Treat proposed nonideal effects as hypotheses supported by evidence.



Arrows point toward dependent results. Planning guide; experimental work remains pending.